

Hydrogen (H₂)

A flammable, colorless, odorless compressed gas at high pressure.

GRADES & SPECIFICATIONS

Grade	% Purity	O ₂	H ₂ O	THC	CO	CO ₂	Ar	N ₂
Research 6.0 ⁽¹⁾	99.9999	0.5	0.5	0.1	0.1	0.1	0.5	0.5
Ultra-Pure Carrier (UPC)	99.9995	1	1	0.5	1 ⁽²⁾	1 ⁽²⁾	—	3
Semiconductor 5.0	99.999	1	2	0.5	1 ⁽²⁾	1 ⁽²⁾	—	5
Ultra High Purity (UHP) 5.0	99.999	1	2	0.5	1 ⁽²⁾	1 ⁽²⁾	—	5
High Purity (HP)	99.995	4	—	—	—	—	—	3

Concentrations given are ppm by volume unless otherwise specified.

(1) Total of impurities listed shall not exceed 1 ppm. (2) CO + CO₂ shall not exceed 1 ppm.

Individual or Batch Certificate of Analysis available upon request.

ORDERING INFORMATION

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Research 6.0	300	260	350	HY-RES-300	2,400	Single / 2-stage · 0–100
	200	216	350	HY-RES-200	2,000	
	80	85	350	HY-RES-80	2,000	
	35	34	350	HY-RES-35	2,000	
Ultra-Pure Carrier (UPC)	300	260	350	HY-UPC-300	2,400	Single-stage
	200	216	350	HY-UPC-200	2,000	
	80	85	350	HY-UPC-80	2,000	
	35	34	350	HY-UPC-35	2,000	
Semiconductor 5.0	300	260	350	HY-SEMI-300	2,400	2-stage
	200	216	350	HY-SEMI-200	2,000	

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Hydrogen (H₂)

Continued — Ordering Information & Technical Data

— ORDERING INFORMATION (CONT.)

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Ultra High Purity (UHP) 5.0	300	260	350	HY-UHP-300	2,400	2-stage
	200	216	350	HY-UHP-200	2,000	
	80	85	350	HY-UHP-80	2,000	
	35	34	350	HY-UHP-35	2,000	
High Purity (HP)	6K	453	703	HY-HP-6K	6,000	0–4,000
	3K	318	695	HY-HP-3K	3,500	0–6,000

— TECHNICAL DATA & SHIPPING INFORMATION

Molecular Weight	2.01588 g/mol	U.S. DOT Name	Hydrogen, Compressed
Specific Volume	191.9 cf/lb @ 70°F & 1 ATM	ID Number	UN 1049
Boiling Point	–252.9 °C	DOT Hazard Class	2.1 — Flammable Gas
Flammability in Air	Flammable — 4.0–75% in Air	CAS Registry Number	1333-74-0

WSM offers a wide range of gas supply modes, including high-pressure cylinders, liquid cylinders (dewars), MicroBulk and bulk. Your WSM specialty gas representative can help you select the option best suited to your operation, including automatic changeover systems for uninterrupted supply.

WSM Quality Policy. The purpose of the WSM Quality System is to continually improve our manufacturing and related processes to provide our customers with the highest product purity, consistency, and service.